

Circumferential multi-contrast mapping of the aqueous outflow pathway in human eyes facilitated by circular-scan optical coherence tomography angiography

ZHAOYU GONG,¹  YAPING SHI,¹ MATTHEW D. HOLLIMAN,²
JINGJIANG XU,¹ YI ZHANG,¹  AND RUIKANG K. WANG^{1,3,*} 

¹Department of Bioengineering, University of Washington, Seattle, WA 98105, USA

²Biological Sciences Collegiate Division, University of Chicago, Chicago, IL 60637, USA

³Department of Ophthalmology, University of Washington, Seattle, WA 98109, USA

*wangrk@uw.edu

Abstract: The corneoscleral limbus contains essential aqueous channels and veins that form the primary aqueous outflow pathway, making it a critical focus of glaucoma research. Optical coherence tomography (OCT) has emerged as a powerful tool for investigating this pathway, but conventional imaging is typically performed in segments and restricted to a single functional contrast. Such approaches are limited in evaluating the overall condition of the outflow system, which contributes circumferentially asymmetrically to the aqueous outflow and is composed of multiple tissues requiring different OCT functional extensions. To address these limitations, we propose a circumferential, multi-contrast mapping framework for in vivo analysis of limbal vasculature and aqueous outflow structures/functions. The workflow begins with a rapid panoramic OCT angiography scan using a circular-scan anterior segment OCT (CircAS-OCT) system to generate a global limbal reference map. This reference is then used to guide the stitching of sectional scans acquired from high-resolution OCT devices and protocols. In two normal subjects, the method successfully enabled circumferential, multi-contrast mapping of selected outflow-related structures and processes, including depth-resolved limbal vasculature, collector channel orifice positions, trabecular meshwork regions, and trabecular motion strength, with results aligning well with established anatomical knowledge. By combining circumferential coverage and cross-function integration, our approach provides a versatile imaging platform with strong potential for comprehensive assessment of the aqueous outflow pathway, supporting diagnostic and surgical strategies in glaucoma management.

© 2026 Optica Publishing Group under the terms of the [Optica Open Access Publishing Agreement](#)

1. Introduction

The corneo-scleral limbus is a highly vascularized region that contains the primary aqueous outflow pathway, which regulates intraocular pressure (IOP) by draining aqueous humor from the anterior chamber into the venous system [1]. This pathway begins at the trabecular meshwork (TM), passes through Schlemm's canal (SC), and exits via collector channels (CC), a deep scleral plexus, and aqueous veins before merging into episcleral veins [2]. Malfunction of this drainage system can elevate IOP, leading to glaucoma characterized by optic neuropathy and vision loss [3]. Importantly, both structural and functional studies have shown that the aqueous outflow pathway is not circumferentially symmetric. Specifically, the outflow through TM tends to be stronger in the inferonasal quadrant [4–6]. Also, pathological deterioration of outflow is often spatially localized rather than uniform distributed, and such focal dysfunction is thought to contribute to glaucoma onset and progression. These findings underscore the need for circumferential assessment of limbal structures and vasculature to provide an unbiased and comprehensive evaluation of outflow system functionality.

Optical coherence tomography (OCT) has emerged as a powerful tool for ophthalmic imaging because of its rapid, non-invasive, three-dimensional capability and high axial resolution ranging from several to tens of micrometers [7–9]. OCT has been applied to study a wide range of outflow-related structures and functions, including Schlemm’s canal morphology [10–12], trabecular meshwork structure [13] and motion [14,15], scleral and conjunctival vasculature [16,17], and tissue birefringence [18], with many of these studies reporting correlations with glaucoma [10,19–21]. However, these measurements often require high spatial or temporal resolution, restricting the imaging field of view, for example, measurements of SC morphology [11,12] and TM dynamics [15]. As a result, most existing OCT methods provide only segmental views, limiting the ability to capture circumferential heterogeneity.

To extend the field of view, researchers have explored wide-field or stitched imaging strategies. OCT angiography (OCTA) can visualize the entire limbal vasculature with an ultrawide-field scan [22,23]. However, in practice, eyelid obstruction often hinders synchronous imaging of both the superior and inferior quadrants. Stitching has therefore become a common approach, typically relying on geometry- or feature-based methods. Geometry-based stitching, which maps different scan perspectives onto a pre-calibrated spherical eye model, has been applied successfully in enucleated [24] and anesthetized eyes [25,26] but is unreliable in awake human subjects due to inter-scan head movements. Feature-based stitching uses intrinsic landmarks such as vasculature [12,15,17,27], palisades of Vogt [12], or external anterior-segment camera images [11]. While effective, reported stitching strategies have been modality-specific, aligning images of a single tissue type (e.g., SC, TM, or vasculature). Given that the aqueous outflow system comprises multiple tissues, each requiring different OCT functional extensions for optimal visualization, a cross-imaging-contrast data integration strategy is necessary to achieve a comprehensive analysis.

In this study, we introduce a robust framework for noninvasive, circumferential, and multi-contrast mapping framework of the aqueous outflow pathway in human eyes. Vascular information is employed as the stitching feature across various imaging contrasts. First, we present a circular-scan anterior-segment OCT (CircAS-OCT) system that enables rapid, single-shot imaging of panoramic limbal vasculature, supported by a dedicated optical design that mitigates defocus. Second, we demonstrate cross-imaging-contrast stitching using the circumferential vasculature map as a global reference, achieving the integration of selected outflow-related structures and processes, including high-resolution and depth-resolved angiography, collector channel orifice (CCO) localization, TM area delineation, and TM motion quantification from different imaging protocols and devices.

2. Materials and methods

2.1. Experimental setup

The CircAS-OCT system employed a 600 kHz MEMS-VCSEL swept laser source (Praevium Research Inc.), with a center wavelength of 1310 nm and an optical bandwidth of 20 nm, providing a theoretical axial resolution of approximately 29.6 μm in tissue (assuming a refractive index of 1.4). The system setup has been detailed in our previous publications [28], and the sample arm was re-designed specifically for the circumferential imaging of the anterior segment.

In conventional anterior-segment OCT, the sample arm is typically designed for telecentric scanning, where the chief rays across the field are parallel to the optical axis. However, due to the strong curvature of the corneoscleral shell, telecentric beams strike the surface at steep angles, restricting the effective depth of focus and inducing defocus artifacts in out-of-plane regions. To address this, the CircAS-OCT employs a pericentric scanning geometry, in which the chief rays virtually converge rather than propagate in parallel, thereby reducing the angle of incidence and alleviating defocus artifacts. In theory, convergence toward the nodal point of the eye would further reduce the angle of incidence toward zero, but at the cost of substantially larger optics and

severe off-axis aberrations. Here, we select the posterior pole of the eyeball as the convergence point as a practical compromise.

The schematic of the CircAS-OCT sample arm is illustrated in Fig. 1(a). The optical path consists of a group of four lenses (L0–L3), designed to produce pericentric chief rays incident on the anterior segment of the eye. The laser beam exiting the galvanometer scanners is initially focused by lens L0. Lens L1 is positioned at the focal plane of L0 to modulate the direction of the chief rays. A subsequent lens, L2, is placed near the pupil plane of the sample arm to collimate the beam without altering the chief ray trajectory. Finally, lens L3 focuses the beam onto the anterior segment, with the chief rays virtually converging at the back of the eyeball. In the physical implementation, depicted in the 2D layout of Fig. 1(c), we employed a telecentric scanning lens (LSM04, Thorlabs Inc.) as L0, and three achromatic doublets for L1 through L3 (L1: AC254-030-C, L2: AC254-060-C, L3: AC508-075-C, Thorlabs Inc.). The sample arm also includes a fixation target (indicated by green beams in Fig. 1(c)) and an integrated RGB camera (blue beams in Fig. 1(c)) to monitor iris position during imaging.

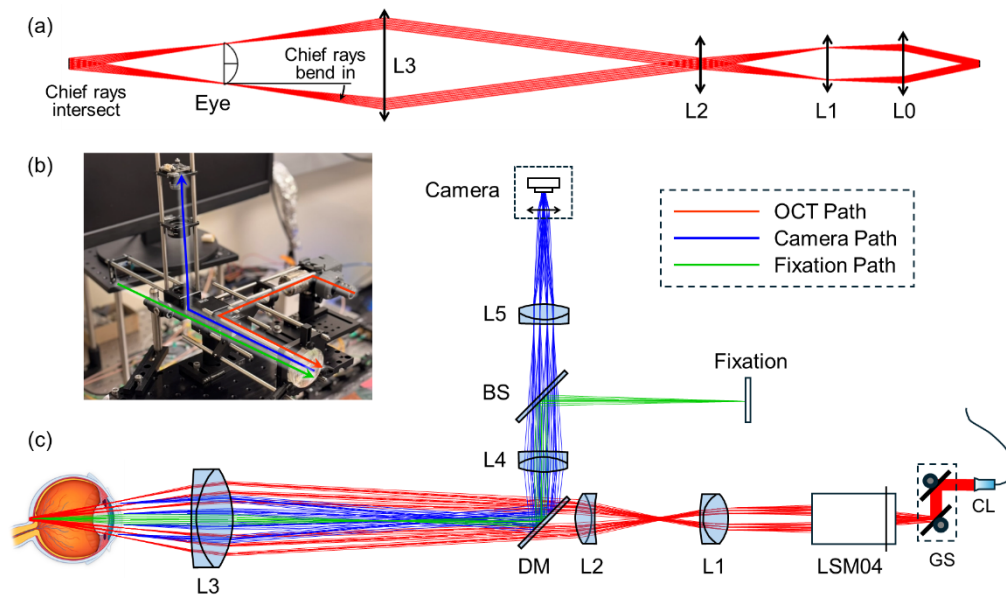


Fig. 1. The design of the sample arm of the CircAS-OCT system. (a) The schematic of the OCT sample arm illustrated using ideal lenses. (b) The photo of the sample arm. (c) The cross-sectional layout of the optical configurations. Red, blue, green beams correspond to the OCT, Iris camera, and Fixation target paths, respectively. L1~L5: achromatic doublets, DM: dichroic mirror, BS: beam splitter, GS: galvanometer scanner, CL: collimator.

2.2. Data acquisition

2.2.1. Subjects and Imaging procedures

Two healthy subjects with no history of ocular or systemic disease were recruited for this study. Only the right eye of each participant was included. The study adhered to the tenets of the Declaration of Helsinki and complied with the Health Insurance Portability and Accountability Act (HIPAA). Ethical approval was obtained from the Institutional Review Board of the University of Washington, and written informed consent was obtained from all participants prior to imaging.

Each subject first underwent imaging with the proposed CircAS-OCT system to acquire a single-shot, circumferential map of the limbal vasculature. Following this, sectional scans were

collected to demonstrate multi-contrast integration: (1) volumetric trabecular meshwork (TM) motion imaging [15] implemented on the CircAS-OCT system, and (2) high-resolution OCTA using a commercial anterior segment OCT device. These datasets were subsequently employed for cross-imaging-contrast stitching, with the CircAS-OCT circumferential vasculature map serving as the global reference.

2.2.2. Single-shot limbal circumferential OCTA

Circumferential imaging of the limbal vasculature was performed using a circular-scan protocol specifically designed to capture the annular region around the corneoscleral limbus. Unlike conventional raster scanning, which covers a rectangular field and includes irrelevant areas such as the pupil center and image corners, the circular protocol focuses exclusively on the limbal circumference. This design, combined with the fast light source, significantly reduces acquisition time and improves subject comfort.

To achieve a circular scanning pattern, the galvanometer mirrors along the x- and y-axes were driven by sinusoidal waveforms of identical frequency but with a constant 90° phase shift, as illustrated in Fig. 2(a). Within each ensemble of unit circular cycles, the amplitudes of the sinusoidal signals remained constant. After completing each ensemble, the amplitude was reduced in a stepwise fashion, as shown in Fig. 2(b), resulting in a gradually contracting scan pattern. This process produced a spiral trajectory that formed an annulus encompassing the entire limbal region (Fig. 2(c)).

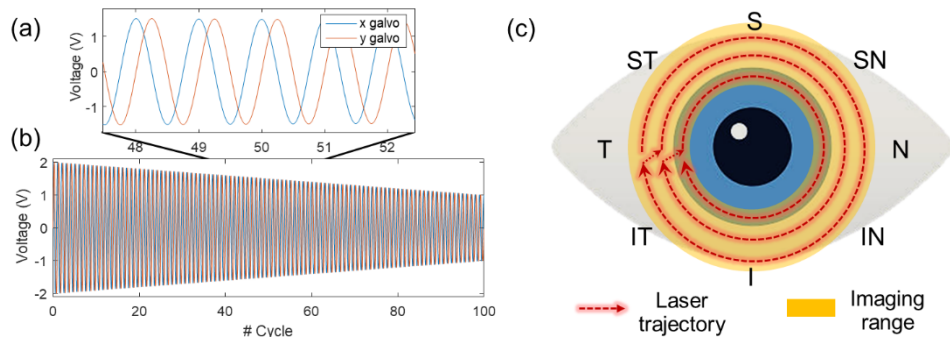


Fig. 2. Scanning protocol for CircAS-OCT. (a) The zoomed-in view of an illustrative scan waveform. (b) The full scan waveform. (c) The schematic of the scanning pattern on the anterior segment of the eye.

The galvanometer scanners moved in equal angular speed mode. Each unit circular cycle consisted of 2000 angular steps, i.e., every B-scan contained 2000 A-scans. To extract blood flow signals, each circular cycle was repeated four times at each radial position. A complete volumetric scan comprised 800 unit circular scans distributed across 200 radial positions, yielding a data volume with dimensions of $1440 \times 2000 \times 200 \times 4$ (pixels $\times$ A-scans $\times$ radial steps $\times$ repeats). The annulus spanned inner and outer radii of 4.25 mm and 8.5 mm, respectively, resulting in a constant radial sampling density of $21.25 \mu\text{m}$. The angular sampling density varied from $13.4 \mu\text{m}$ to $26.7 \mu\text{m}$ across the field.

During the imaging session, the subject was instructed to fixate on the built-in target to maintain gaze along the optical axis of the system. The operator then adjusted the imaging probe until the scanning field was aligned concentrically with the pupil before initiating the scan. The acquisition time for the entire circumferential scan was 2.67 seconds.

2.2.3. High-resolution OCTA

A commercial swept-source OCT system (VG-200D, Intalight Inc., China) was used to capture the high-resolution images of the anterior segment. The system employed a 200 kHz swept laser with a center wavelength of 1050 nm, providing an axial resolution of 5.5 μm and a lateral resolution of 20 μm in tissue. An anterior segment imaging protocol with a raster scanning pattern was employed, using a lateral sampling grid of 384×384 points to cover a $6 \times 6 \text{ mm}^2$ area on the ocular surface. At each cross-sectional location, B-scans were repeated three times to extract blood flow signals. The total acquisition time for each scan was approximately 3.5 seconds.

During imaging, subjects were instructed to gaze obliquely at an external LED fixation target. This positioning ensured that the targeted ocular surface was aligned perpendicular to the optical axis of the objective lens, thereby optimizing the focus across the entire field of view. Eight sections of the limbus were scanned in sequence: supero-temporal (ST), superior (S), supero-nasal (SN), nasal (N), infero-nasal (IN), inferior (I), infero-temporal (IT), and temporal (T). Each section was imaged three times, and the image with the highest quality from each set was selected for further analysis. The total time to capture these images on an individual eye is 20~30 minutes.

2.2.4. Volumetric TM motion imaging

The scanning protocol for capturing TM dynamics followed our previously reported method [15]. In brief, repeated volumetric acquisitions were performed over a $6 \times 2 \text{ mm}^2$ lateral field centered at the corneoscleral limbus for a duration of 5 seconds. This protocol simultaneously captured tissue motion and vasculature information. For each subject, this imaging protocol was repeated four times. Slight offset was introduced after each scan to traverse the scan area across the inferior quadrants, covering a continuous TM band of approximately 5 mm in length. The total time of capturing the inferior TM motion is 5 minutes approximately.

2.3. Data processing

2.3.1. Single-shot circumferential OCTA

OCTA images from the circular scans were generated following standard procedures [29]. The acquired OCT interferograms first underwent background subtraction and optical dispersion compensation [30], and were then Fourier transformed to obtain depth-resolved signals. For each ensemble of repeated B-scans, a subpixel registration algorithm [31] and a phase stabilization algorithm [32] were applied to minimize bulk motion. Notably, the subpixel registration did not include any column shifts, as bulk eye motion does not appear as translational displacement in the circular scan direction. Subsequently, an eigen-decomposition-based angiography algorithm [33] was used to separate static and dynamic components. The separated static and dynamic signals from all ensembles were assembled into the OCT structural and OCTA volumes (Fig. 3(a1)), respectively. The superficial tissue surface was segmented from the structural volume and used to generate *en face* projections of the OCTA volume (Fig. 3(a2)). In this study, a mean intensity projection was performed over a depth slab ranging from 100 to 300 μm below the tissue surface. The resulting angiograms were contrast-enhanced using adaptive histogram equalization (Fig. 3(a3)) and resampled to Cartesian coordinates for visualization (Fig. 3(a4)).

2.3.2. Sectional scans

OCTA volumes acquired from the commercial system were exported as DICOM files. *En face* angiograms were generated by performing maximum intensity projection (MIP) along each A-scan from the ocular surface to the inner boundaries of the cornea and sclera. Depth information was preserved in the projection by encoding the relative distance to the ocular surface with a green–yellow–red colormap. In addition, the projection range was further separated into

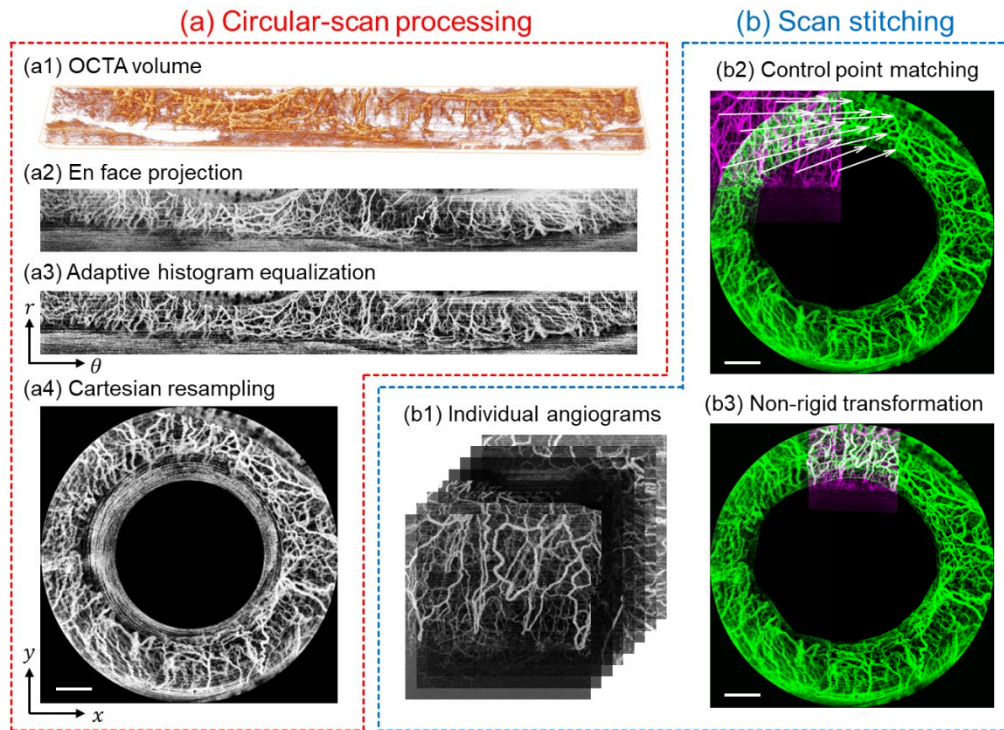


Fig. 3. Illustration of data processing steps. (a1)-(a4) The processing steps of circular-scan OCTA, generating a global reference to stitch local angiograms. (b1)-(b3) The stitching of angiograms from other sectional imaging protocols to the global reference. Scale bars = 2 mm.

superficial, middle, and deep slabs, in which the *en face* angiograms were individually generated, and blood flow projection artifacts in the deeper slabs were removed by a validated algorithm [34]. All B-scans were manually inspected by a certified ophthalmic professional (MDH) to identify collector channel orifice (CCO) locations, which were recorded as corresponding row and column coordinates in the *en face* angiogram. Details about the annotation process is included in Supplement 1, Note 1.

Processing of the volumetric TM motion data followed the approach described in our previous work [15]. Briefly, customized registration algorithms were first applied to minimize eye-motion artifacts across the sequential volumes. Phase shifts and blood flow signals were then extracted from each pair of consecutive volumes. Segmentation of the TM region was achieved by correlating OCT phase dynamics with the cardiac cycle, based on the hypothesis that TM motion is synchronized with the heartbeat. Within the segmented TM mask, motion amplitude was calculated as a function of angular position along the limbal circumference. Flow signals from all volume pairs were subsequently averaged, and *en face* angiograms were generated from the averaged flow data.

2.3.3. Angio-guided cross-imaging-contrast stitching

The angiogram generated from the circular scan served as a global reference for registering and stitching angiograms acquired from other sectional imaging protocols, as illustrated in Fig. 3(b).

To account for complex distortions arising from perspective differences on the approximately spherical ocular surface, we adopted a coarse-to-fine registration strategy to progressively

refine vascular alignment. First, each individual angiogram was coarsely aligned to the reference using a projective transformation fitted to manually selected control points via the MATLAB function `cpselect()` (Fig. 3(b2)). This coarse alignment was then refined with non-rigid registration implemented using the MATLAB function `imregdform()` (Fig. 3(b3)). Following registration, all angiograms were mapped to their corresponding circumferential positions along the limbus, and overlapping regions were fused using a linear blending method.

The same transformation was subsequently applied to fuse additional biometric data. Specifically, CCO coordinates from high-resolution OCTA scans, as well as TM segmentations and motion amplitude maps from volumetric motion imaging, were transformed into the circumferential coordinate system. This process yielded an integrated, multi-contrast representation of the aqueous outflow pathway.

3. Results

3.1. CircAS-OCT system characteristics

Figure 4 summarizes the evaluation of system imaging performance of our CircAS-OCT. The first two columns present optical simulations conducted in Ansys Zemax OpticStudio (Ansys Inc., USA). The imaging plane is set to have a radius of 11.5 mm – matching the biometric of human limbus [35]. Spot diagrams at radii of 3 mm, 6 mm, and 9 mm are shown in Figs. 4(a1)–4(a3).

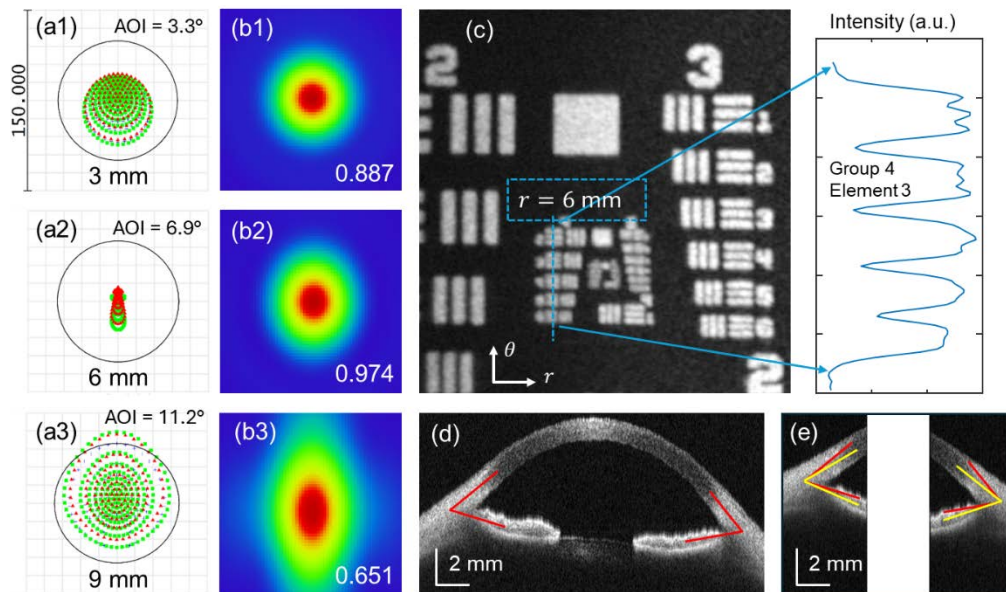


Fig. 4. Imaging performance of the CircAS-OCT system. (a1)-(a3) Spot diagrams at field radii of 3 mm, 6 mm, and 9 mm. Red, green, and blue markers correspond to wavelengths of 1300 nm, 1310 nm, and 1320 nm, respectively. Black circles denote the Airy disks, and the angles of incidence (AOI) are shown in the upper-right corners. (b1)-(b3) Point spread functions at the same field positions, with Strehl ratios displayed in the lower-right corners. (c) USAF resolution target imaged at a 6-mm radius. The graph on the right shows the line profile of group 4, where element 3 is the finest resolvable feature. (d) Cross-sectional anterior segment image acquired with telecentric scanning optics. The anterior chamber angle skeletons are outlined in red. (e) Cross-sectional image obtained with the CircAS-OCT circular scanning protocol. The corresponding skeletons are outlined in yellow and compared against those from the telecentric configuration.

The incidence angles, displayed in the upper-right corner of each panel, increase with field size, illustrating the effect of the pericentric design. Corresponding point spread functions (Figs. 4(b1)–(4b3)) demonstrate stable spot sizes across the field, with only mild radial blurring observed at the periphery compared to the central region.

Figure 4(c) reports experimental measurements of lateral resolution using a USAF resolution target. At a 6-mm radius, the system resolved group 4, element 3, corresponding to a lateral resolution of 49.6 μm .

Finally, the impact of the pericentric design on incidence angles was validated with OCT B-scans. Figures 4(d) and 4(e) compare cross-sectional images of the anterior segment acquired with a telecentric lens (LSM05, Thorlabs Inc.) and our pericentric beam profiles, respectively. The anterior chamber angle boundaries, highlighted by red (telecentric) and yellow (pericentric) skeletons, display a constant rotational offset of approximately nine degrees ($\sim 9^\circ$) in Fig. 4(e), confirming the reduced incidence angles achieved with the pericentric configuration.

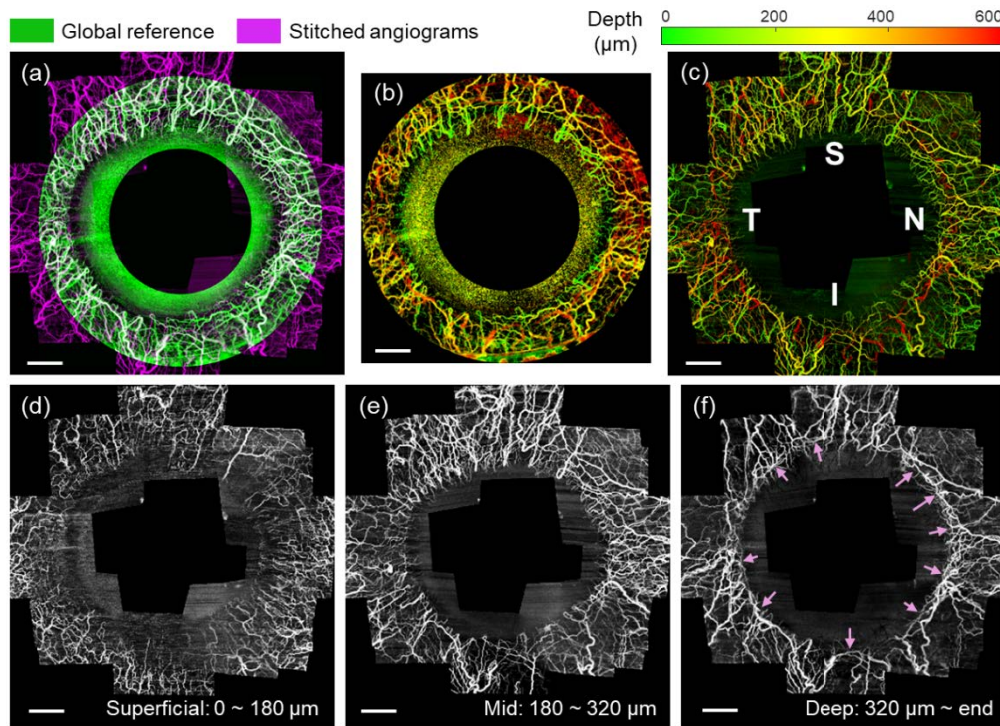


Fig. 5. Representative images of the circumferential limbal vasculature. (a) Overlay of stitched high-resolution angiograms from the commercial system registered to the global reference. (b) Single-shot circumferential vascular global reference acquired with the CircAS-OCT system, displayed with a depth-encoded color map. (c) Stitched high-resolution circumferential vascular map with depth-encoded color representation. (d) Superficial vascular slab (0–180 μm from the ocular surface), highlighting predominantly conjunctival vessels and corneal arcades. (e) Middle vascular slab (180–320 μm), corresponding mainly to episcleral vessels. (f) Deep vascular slab (320 μm to the posterior stroma), containing the deep scleral plexus. The magenta arrows highlight the presence of circumferential vascular compartment (CVC). Scale bars = 2 mm.

3.2. Circumferential vasculature imaging

Imaging and stitching results for the circumferential vasculature are summarized in Fig. 5. Figure 5(a) shows the overlay of stitched high-resolution vascular images (magenta) with the global reference angiogram from the CircAS-OCT system (green). Vessel skeletons from both datasets exhibited excellent spatial correspondence, with co-registered vessels appearing white in the overlay. Figures 5(b) and 5(c) present the global reference (Fig. 5(b)) and the stitched high-resolution angiograms (Fig. 5(c)) individually, each displayed with the same depth-encoded colormap. As expected, vessels in the high-resolution angiograms appeared finer and more sharply delineated, reflecting the difference in spatial resolutions of the commercial system compared to CircAS-OCT.

Figures 5(d)–5(f) show *en face* angiograms from three depth-resolved vascular slabs derived from the high-resolution dataset. The superficial slab (0–180 μm from the ocular surface; Fig. 5(d)) predominantly contains conjunctival vessels and corneal arcades, which appear as the thinnest vessels among the three slabs. The middle slab (180–320 μm ; Fig. 5(e)) corresponds mainly to episcleral vasculature, characterized by radially oriented vessels of larger caliber than those in the superficial slab. The deep slab (320 μm to the posterior scleral stroma; Fig. 5(f)) contains more circumferentially oriented vasculature than other slabs, including the circumferential vascular compartment (CVC), as pointed by the pink arrows in Fig. 5(f). The CVC lies concentrically with SC but at a larger radius around the limbus, serving as a major downstream structure of SC and playing a key role in aqueous humor outflow.

3.3. Multi-contrast data integration

Figure 6 summarizes the results of multi-contrast data integration from two healthy subjects. The combined datasets include: (1) circumferential depth-encoded vascular maps; (2) collector channel orifice (CCO) positions, marked by white triangles (Figs. 6(a) and 6(b)); (3) trabecular meshwork (TM) regions in the inferior quadrants, delineated by magenta polygons (Figs. 6(a) and 6(b)); and (4) TM motion amplitudes, quantified as peak-to-trough velocities across all recorded cardiac cycles at different angular positions (Figs. 6(c) and 6(d)). To provide B-scan-level evidence of the TM motion measurements obtained with the CircAS-OCT system, representative examples are presented in the [Supplement 1](#), Note 2.

The integrated maps clearly highlight the spatial relationships of aqueous outflow structures. First, CCOs were distributed around the limbal circumference with noticeable variability in spacing—some appearing in close clusters, while others were separated by wider intervals. The CCOs also aligned concentrically with the circumferential vascular compartments (CVCs) in the deep scleral plexus (purple arrows in Figs. 6(a), 6(b)), but were generally located at slightly smaller or comparable radii, aligning with anatomic studies [36]. In addition, CCOs were localized within the TM areas, consistent with their anatomical position below Schlemm's canal. Lastly, TM motion amplitudes, measured in the inferior quadrants, revealed distinct spatial patterns—localized fluctuations superimposed on broader, gradually varying trends—consistent with our previous observations [15].

4. Discussion

In this work, we introduced a methodology for *in vivo* multi-contrast analysis of the aqueous outflow pathway. In the preliminary human study with two normal subjects, we demonstrated the feasibility of co-registering depth-resolved limbal vasculature, CCO positions, TM areas, and TM motion amplitudes obtained from different imaging protocols and devices. This proof-of-concept illustrates how multiple complementary datasets can be integrated to provide a more comprehensive and circumferential characterization of aqueous outflow.

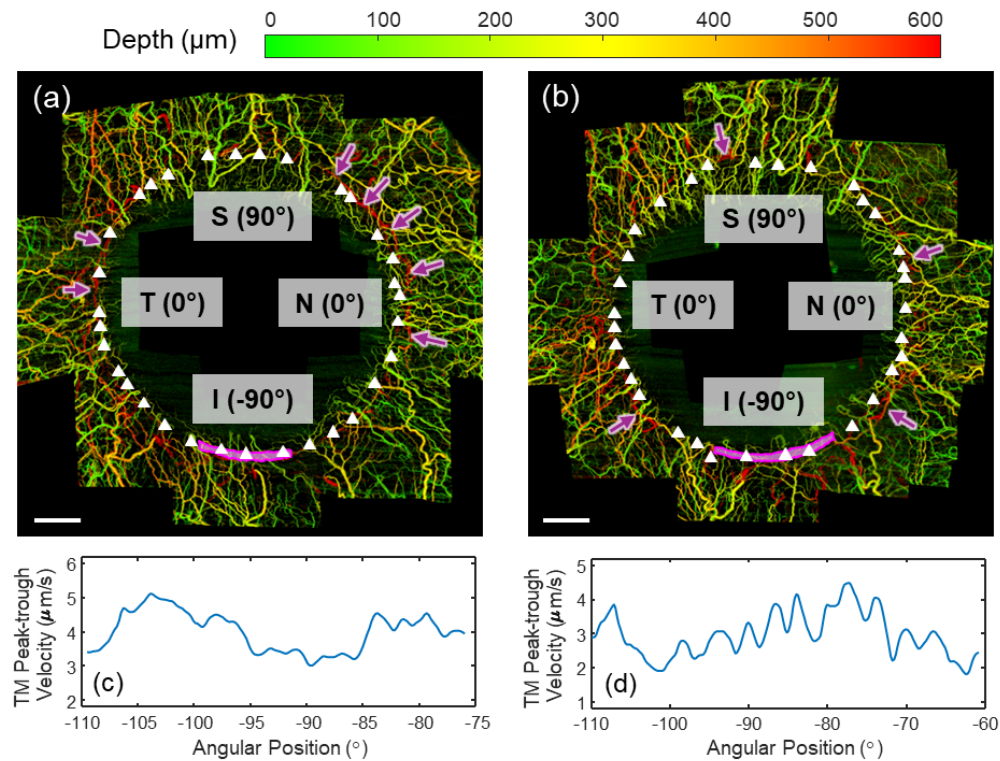


Fig. 6. Multi-contrast data integration results of two normal subjects. (a) and (b) are the co-registered limbal vasculature with CCO positions (white triangular markers) and segmented TM areas at inferior quadrants (magenta polygons). Purple arrows point at the locations of CVCs. Scale bars = 2 mm. (c) and (d) are the TM motion amplitudes quantified by peak-trough velocities at different angular positions.

We selected vascular information as the guiding feature for cross-imaging-contrast registration. This choice is supported by prior evidence that OCTA-derived vascular morphology exhibits high repeatability in both the anterior segment [37] and retina [38], with large vessels demonstrating especially robust inter-visit reproducibility [39]. Such stability makes vasculature a reliable anchor for stitching datasets acquired from different modalities or at different time points. Nevertheless, certain limitations should be noted. Vessel diameters and the visibility of small-caliber vessels can vary between systems due to differences in spatial resolution, repetition rates, and OCTA contrast sensitivity (as observed in Figs. 5(b)–5(c)). These discrepancies occasionally caused the automated registration algorithm to fail to converge. To improve robustness, we employed manual selection of a few control points to initialize projective alignment, which significantly increased the success rate of subsequent non-rigid refinement. This manual step can be replaced by automated feature-matching in future implementations to handle larger datasets.

The reliability of cross-imaging-contrast stitching also depends on the quality of the global reference. Specifically in the single-shot acquisition of circumferential reference, challenges include defocus artifacts introduced by the pronounced curvature of the ocular surface and the trade-off between scanning speed and field of view. The CircAS-OCT system was specifically designed to address these issues. By employing a state-of-the-art 600 kHz MEMS-VCSEL source and adopting a circular-scan pattern, we reduced scan time to 2.67 seconds while maintaining full circumferential coverage of the limbus. Also, the design of an annular scan pattern skips

unnecessary central and corner regions, reducing the sampling area by $\sim 40\%$ compared to the traditional rectangular raster-scan pattern. Therefore, the CircAS-OCT system plus the annular scan pattern collectively improves subject comfort and compliance. Currently, the equal-angular speed strategy leads to higher angular sampling density near the center, leading to unwanted oversampling. Further scan acceleration might include unifying the sampling density across all radii.

Another strength of CircAS-OCT lies in its pericentric beam design, which reduces angles of incidence and adapts the focal plane to the curved limbal surface. This minimizes defocus artifacts and improves uniformity across the field. Similar strategies have been employed for high-quality corneal imaging [40–42]. Some even achieved zero AOI by converging the scanning chief rays at the posterior nodal point. However, imaging the perilimbal area requires a larger imaging field compared to corneal imaging, which can considerably increase the aperture of the optics, as reported in the commercial system Pentacam® Cornea OCT (OCULUS Optikgeräte GmbH, Germany) [43], and condenser-lens design [40,44]. However, the off-axis aberration is extremely hard to control, especially the distortion and the chromatic aberration. Our design instead employs a more compact and balanced design by using a 2-inch achromatic doublet, achieving both angle-of-incidence reduction (Fig. 4(e)) and control of off-axis aberrations (Figs. 4(a3), (4b3)). With this design, it should be acknowledged that eyes with extreme white-to-white diameters or altered peripheral scleral curvature may still experience reduced resolution at the outer scan radius. Thoroughly addressing this defocus may require implementing dynamic focusing to track the limbal topography [45].

One limitation lies in the acquisition of single-shot circumferential OCTA, which requires manual eyelid retraction to avoid shadowing. While the short acquisition time alleviates this burden in healthy volunteers, clinical translation may be challenging in patients with unstable fixation or reduced tolerance. Two strategies may mitigate this: (1) topical anesthesia to reduce eye sensitivity, or (2) dividing the acquisition into two semi-circular scans with independent eyelid retraction, followed by montaging to generate a full circumferential map.

The spatial relationships of aqueous outflow structures illustrated in Fig. 6 are consistent with established anatomical findings and demonstrate their feasibility to be visualized *in vivo*. As a proof-of-concept, our study was limited to a small sample size of normal eyes. With a larger cohort, these spatial relationships could be quantified and analyzed statistically, potentially revealing quadrant-specific differences or other physiological patterns at the population level.

The multi-contrast mapping capabilities are demonstrated only on partial outflow-related structures and processes, but the proposed platform is inherently extensible. Any OCT protocol that incorporates repeated measurements can, in principle, generate vasculature information suitable for stitching. Examples include OCTA [46], OCT velocimetry [47,48], polarization-sensitive OCT with repetition to reduce the uncertainty [49], volume-repeat phase-sensitive OCT [50–53], dynamic OCT [54,55]. Even single-repeat protocols may yield large-vessel maps via deep-learning-driven tissue-blood classification [56,57]. This broad compatibility positions CircAS-OCT as a flexible hub for integrating diverse functional and structural metrics of the outflow system, including Schlemm's canal morphology [12,25], optical attenuation coefficients [58], and tissue birefringence [18].

Future clinical studies are warranted to validate this platform in glaucoma patients and to evaluate its diagnostic utility. In particular, circumferential mapping may help identify subtle, localized dysfunction in aqueous outflow that is not captured by segmental imaging. Moreover, multi-contrast integration could reveal spatial correlations among different outflow metrics—such as IOP, TM motion amplitude, SC morphology, and vascular patterns—providing new insights into the physiology of outflow regulation. Together, these capabilities may improve the sensitivity of glaucoma diagnosis and broaden our understanding of aqueous humor dynamics in the healthy and diseased.

5. Conclusion

In this study, we introduced a robust framework for noninvasive, circumferential, and multi-contrast mapping framework for the aqueous outflow pathway in human eyes. The workflow is built around a novel CircAS-OCT system, which enables rapid and focus-optimized acquisition of a panoramic OCT angiogram of the limbal circumference, combined with a customized registration framework for stitching angiograms acquired from other high resolution OCT systems. This approach successfully captured fine vascular structures and depth-resolved details across the entire limbal circumference. Moreover, the same transformation framework was extended to register additional outflow-related features, such as positions of CCOs, TM areas and TM motion amplitudes, demonstrating the flexibility and extensibility of the method. Together, these capabilities establish CircAS-OCT as a versatile imaging platform that unites wide-field coverage, high-resolution detail, and multi-contrast data integration. Such advances may open new opportunities for comprehensive anterior segment assessment and hold strong potential to support both research and clinical strategies for glaucoma diagnosis and management.

Funding. Kren Engineering-based Medicine Initiative (KEMi) Fund; George and Martina Kren Research Fund in Bioengineering, University of Washington Department of Bioengineering.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but can be obtained from the authors upon reasonable request.

Supplemental document. See [Supplement 1](#) for supporting content.

References

1. M. Johnstone, C. Xin, J. Tan, *et al.*, "Aqueous outflow regulation – 21st century concepts," *Prog. Retinal Eye Res.* **83**, 100917 (2021).
2. T. Carreon, E. van der Merwe, R. L. Fellman, *et al.*, "Aqueous outflow - A continuum from trabecular meshwork to episcleral veins," *Prog. Retinal Eye Res.* **57**, 108–133 (2017).
3. The AGIS Investigators, "The advanced glaucoma intervention study (AGIS): 7. the relationship between control of intraocular pressure and visual field deterioration," *Am. J. Ophthalmol.* **130**(4), 429–440 (2000).
4. R. R. Allingham, A. W. de Kater, and R. C. Ethier, "Schlemm's canal and primary open angle glaucoma: correlation between Schlemm's canal dimensions and outflow facility," *Exp. Eye Res.* **62**(1), 101–110 (1996).
5. M. Johnstone, A. Jamil, and E. Martin, "Aqueous veins and open angle glaucoma," in *The Glaucoma Book: A Practical, Evidence-Based Approach to Patient Care*, P. N. Schacknow and J. R. Samples, eds. (Springer New York, 2010), pp. 65–78.
6. J. A. Vranka, J. M. Bradley, Y.-F. Yang, *et al.*, "Mapping molecular differences and extracellular matrix gene expression in segmental outflow pathways of the human ocular trabecular meshwork," *PLoS One* **10**(3), e0122483 (2015).
7. P. H. Tomlins and R. K. Wang, "Theory, developments and applications of optical coherence tomography," *J. Phys. D: Appl. Phys.* **38**(15), 2519–2535 (2005).
8. F. Zhang, K. Kovalick, A. Raghavendra, *et al.*, "In vivo imaging of human retinal ganglion cells using optical coherence tomography without adaptive optics," *Biomed. Opt. Express* **15**(8), 4675–4688 (2024).
9. K. Neuhaus, S. Khan, O. Thaware, *et al.*, "Real-time line-field optical coherence tomography for cellular resolution imaging of biological tissue," *Biomed. Opt. Express* **15**(2), 1059–1073 (2024).
10. L. Kagemann, G. Wollstein, H. Ishikawa, *et al.*, "Identification and assessment of Schlemm's Canal by spectral-domain optical coherence tomography," *Invest. Ophthalmol. Visual Sci.* **51**(8), 4054–4059 (2010).
11. S. H. Alex, B. Akram, D. Anna, *et al.*, "Automated circumferential construction of first-order aqueous humor outflow pathways using spectral-domain optical coherence tomography," *J. Biomed. Opt.* **22**(6), 066010 (2017).
12. X. Yao, B. Tan, Y. Ho, *et al.*, "Full circumferential morphological analysis of Schlemm's canal in human eyes using megahertz swept source OCT," *Biomed. Opt. Express* **12**(7), 3865–3877 (2021).
13. A. Carmichael-Martins, T. J. Gast, B. J. King, *et al.*, "Imaging fine structures of the human trabecular meshwork in vivo using a custom design gonioscopy and OCT gonioscopy," *Biomed. Opt. Express* **14**(10), 5267–5281 (2023).
14. P. Li, T. T. Shen, M. Johnstone, *et al.*, "Pulsatile motion of the trabecular meshwork in healthy human subjects quantified by phase-sensitive optical coherence tomography," *Biomed. Opt. Express* **4**(10), 2051–2065 (2013).
15. Z. Gong, Y. Shi, J. Liu, *et al.*, "Volumetric imaging of trabecular meshwork dynamic motion using 600 kHz swept source optical coherence tomography," *Biomed. Opt. Express* **16**(1), 267–281 (2025).
16. P. Li, L. An, R. Reif, *et al.*, "In vivo microstructural and microvascular imaging of the human corneo-scleral limbus using optical coherence tomography," *Biomed. Opt. Express* **2**(11), 3109–3118 (2011).

17. T. Akagi, A. Uji, A. S. Huang, *et al.*, "Conjunctival and intrascleral vasculatures assessed using anterior segment optical coherence tomography angiography in normal eyes," *Am. J. Ophthalmol.* **196**, 1–9 (2018).
18. D. Kasaragod, S. Makita, S. Fukuda, *et al.*, "Bayesian maximum likelihood estimator of phase retardation for quantitative polarization-sensitive optical coherence tomography," *Opt. Express* **22**(13), 16472–16492 (2014).
19. S. Sucijanti, Z. L. Yuan, Y. Liang, *et al.*, "Analysis of Schlemm's canal measurement and evaluation with fourier-domain optical coherence tomography in primary open-angle glaucoma," *Sains Malays.* **53**(9), 3097–3111 (2024).
20. K. Gao, S. Song, M. A. Johnstone, *et al.*, "Reduced pulsatile trabecular meshwork motion in eyes with primary open angle glaucoma using phase-sensitive optical coherence tomography," *Invest. Ophthalmol. Visual Sci.* **61**(14), 21 (2020).
21. Y. Okamoto, T. Akagi, T. Kameda, *et al.*, "Prediction of trabecular meshwork-targeted micro-invasive glaucoma surgery outcomes using anterior segment OCT angiography," *Sci. Rep.* **11**(1), 17850 (2021).
22. A. Fard, Y. Wang, J. Kost, *et al.*, "Ultra-widefield anterior segment OCT angiography for visualization of iris, limbus and sclera," *Invest. Ophthalmol. Visual Sci.* **61**, PB0028 (2020).
23. L. Lucchino, S. Vaccaro, M. Airaldi, *et al.*, "Advanced imaging of corneal neovascularisation with a novel swept-source AS-OCTA system," *Br. J. Ophthalmol.* **110**(1), 25–30 (2026).
24. S. Luo, G. Holland, R. Khazaeinezhad, *et al.*, "Iridocorneal angle imaging of a human donor eye by spectral-domain optical coherence tomography," *Sci. Rep.* **13**(1), 13861 (2023).
25. X. Zhang, L. Beckmann, D. A. Miller, *et al.*, "In vivo imaging of Schlemm's canal and limbal vascular network in mouse using visible-light OCT," *Invest. Ophthalmol. Visual Sci.* **61**(2), 23 (2020).
26. R. Fang, P. Zhang, D. Kim, *et al.*, "Robotic visible-light optical coherence tomography visualizes segmental Schlemm's canal anatomy and segmental pilocarpine response," *Invest. Ophthalmol. Visual Sci.* **66**(2), 47 (2025).
27. L. Ni, W. Zhang, W. Kim, *et al.*, "3D imaging of aqueous veins and surrounding sclera using a dual-wavelength photoacoustic microscopy," *Biomed. Opt. Express* **14**(12), 6291–6300 (2023).
28. Y. Shi, J. Liu, and R. K. Wang, "High-speed, long-range and wide-field OCT for in vivo 3D imaging of the oral cavity achieved by a 600 kHz swept source laser," *Biomed. Opt. Express* **15**(7), 4365–4380 (2024).
29. S. Song, K. Zhou, J. J. Xu, *et al.*, "Development of a clinical prototype of a miniature hand-held optical coherence tomography probe for prematurity and pediatric ophthalmic imaging," *Biomed. Opt. Express* **10**(5), 2383–2398 (2019).
30. R. K. Wang and Z. Ma, "A practical approach to eliminate autocorrelation artefacts for volume-rate spectral domain optical coherence tomography," *Phys. Med. Biol.* **51**(12), 3231–3239 (2006).
31. M. Guizar-Sicairos, S. T. Thurman, and J. R. Fienup, "Efficient subpixel image registration algorithms," *Opt. Lett.* **33**(2), 156–158 (2008).
32. S. Song, J. Xu, S. Men, *et al.*, "Robust numerical phase stabilization for long-range swept-source optical coherence tomography," *J. Biophotonics* **10**(11), 1398–1410 (2017).
33. S. Yousefi, Z. Zhi, and R. K. Wang, "Eigendecomposition-based clutter filtering technique for optical microangiography," *IEEE Trans. Biomed. Eng.* **58**(8), 2316–2323 (2011).
34. A. Zhang, Q. Zhang, and R. K. Wang, "Minimizing projection artifacts for accurate presentation of choroidal neovascularization in OCT micro-angiography," *Biomed. Opt. Express* **6**(10), 4130–4143 (2015).
35. M. J. Hogan, J. A. Alvarado, and J. E. Weddell, *Histology of the Human Eye: An atlas and textbook*, M. J. Hogan, J. A. Alvarado, and J. E. Weddell, eds. (Saunders, 1971).
36. M. Johnstone, C. Xin, T. Acott, *et al.*, "Valve-like outflow system behavior with motion slowing in glaucoma eyes: findings using a minimally invasive glaucoma surgery-migs-like platform and optical coherence tomography imaging," *Front. Med.* **9**, 815866 (2022).
37. S. Cai, F. Zhao, and C. Du, "Repeatability of ocular surface vessel density measurements with optical coherence tomography angiography," *BMC Ophthalmol.* **19**(1), 248 (2019).
38. J. P. Venugopal, H. L. Rao, R. N. Weinreb, *et al.*, "Repeatability of vessel density measurements of optical coherence tomography angiography in normal and glaucoma eyes," *Br. J. Ophthalmol.* **102**(3), 352–357 (2018).
39. J. Hong, M. Ke, B. Tan, *et al.*, "Effect of vessel enhancement filters on the repeatability of measurements obtained from widefield swept-source optical coherence tomography angiography," *Sci. Rep.* **10**(1), 22179 (2020).
40. F. Beer, A. Wartak, R. Haindl, *et al.*, "Conical scan pattern for enhanced visualization of the human cornea using polarization-sensitive OCT," *Biomed. Opt. Express* **8**(6), 2906–2923 (2017).
41. V. Mazlin, K. Irsch, M. Paques, *et al.*, "Curved-field optical coherence tomography: large-field imaging of human corneal cells and nerves," *Optica* **7**(8), 872–880 (2020).
42. M. Ruggeri, F. P. Giuffrida, N. L. Vy Truong, *et al.*, "Wide-field self-referenced optical coherence tomography imaging of the corneal microlayers," *Opt. Lett.* **50**(4), 1204–1207 (2025).
43. R. Ambrósio Jr, L. P. G. Esporcatte, K. A. de Carvalho, *et al.*, "Combined rotating ultra-high-resolution spectral domain OCT and Scheimpflug imaging for in vivo corneal optical biopsy," *Diagnostics* **14**(13), 1455 (2024).
44. F. Beer, R. P. Patil, A. Sinha-Roy, *et al.*, "Ultrahigh resolution polarization sensitive optical coherence tomography of the human cornea with conical scanning pattern and variable dispersion compensation," *Appl. Sci.* **9**(20), 4245 (2019).
45. J. Liu, Y. Shi, Z. Gong, *et al.*, "Adaptive contour-tracking to aid wide-field swept-source optical coherence tomography imaging of large objects with uneven surface topology," *Biomed. Opt. Express* **15**(8), 4891–4908 (2024).

46. C.-L. Chen and R. K. Wang, "Optical coherence tomography based angiography [Invited]," *Biomed. Opt. Express* **8**(2), 1056–1082 (2017).
47. K. W. Ruikang, Z. Qinqin, L. Yuandong, *et al.*, "Optical coherence tomography angiography-based capillary velocimetry," *J. Biomed. Opt.* **22**(6), 066008 (2017).
48. R. Maltais-Tariant, R. I. Becerra-Deana, S. Brais-Brunet, *et al.*, "Accurate flow speed measurement through correlation ratios using a mode-selective photonic lantern in optical coherence tomography," *Biomed. Opt. Express* **16**(8), 3395–3414 (2025).
49. Y. Shi, J. Liu, Z. Gong, *et al.*, "Minimize flow-induced uncertainty in polarization sensitive optical coherence tomography imaging using eigen decomposition," *Opt. Lett.* **49**(21), 6029–6032 (2024).
50. Z. Gong, Y. Shi, J. Liu, *et al.*, "Light-adapted flicker-optoretinography based on raster-scan optical coherence tomography towards clinical translation," *Biomed. Opt. Express* **15**(10), 6036–6051 (2024).
51. Z. Liu, F. Zhang, K. Zucca, *et al.*, "Ultrahigh-speed multimodal adaptive optics system for microscopic structural and functional imaging of the human retina," *Biomed. Opt. Express* **13**(11), 5860–5878 (2022).
52. B. Tan, H. Li, Y. Zhuo, *et al.*, "Light-evoked deformations in rod photoreceptors, pigment epithelium and subretinal space revealed by prolonged and multilayered optoretinography," *Nat. Commun.* **15**(1), 5156 (2024).
53. R. K. Wang, S. Kirkpatrick, and M. Hinds, "Phase-sensitive optical coherence elastography for mapping tissue microstrains in real time," *Appl. Phys. Lett.* **90**(16), 164105 (2007).
54. K. Fei, Z. Luo, Y. Chen, *et al.*, "Cellular structural and functional imaging of donor and pathological corneas with label-free dual-mode full-field optical coherence tomography," *Biomed. Opt. Express* **15**(6), 3869–3888 (2024).
55. N. Heldt, T. Monfort, R. Morishita, *et al.*, "Guide to dynamic OCT data analysis," *Biomed. Opt. Express* **16**(11), 4851–4870 (2025).
56. C. S. Lee, A. J. Tying, Y. Wu, *et al.*, "Generating retinal flow maps from structural optical coherence tomography with artificial intelligence," *Sci. Rep.* **9**(1), 5694 (2019).
57. M. Chen, Z. Lu, P. Tang, *et al.*, "Investigating the impact of different time delays between OCT signal and k-clock signal on the structural and vascular imaging in SS-OCT," *Biomed. Opt. Express* **16**(3), 1225–1239 (2025).
58. Y. Zhang, Z. Gong, Y. Shi, *et al.*, "Quantitative assessment of retinal attenuation and backscattering in OCT imaging using iterative layer-based analysis," *Biomed. Opt. Express* **16**(7), 2808–2823 (2025).